practice question:- 001

Aim

To study and implement the **WHERE clause** in SQL.

Problem Statement

Create a database table named **Students** to store student details and use the WHERE clause to filter specific records.

Table: Students

StudentID	Name	Age	Course	Marks
101	Rahul	20	BCA	85
102	Priya	21	BBA	78
103	Amit	22	BCA	90
104	Neha	20	BCom	72
105	Arjun	21	BCA	88

Tasks to Perform

(a) Create the Students table.
(b) Insert the given records.
(c) Display students who belong to BCA course.
(d) Display students who scored more than 80 marks.
(e) Display students whose age is 21.

Solution

Step 1: Create Table

```
CREATE TABLE Students (
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50),
    Age INT,
    Course VARCHAR(20),
    Marks INT
);
```

Step 2: Insert Records

```
INSERT INTO Students VALUES
(101,'Rahul',20,'BCA',85),
(102,'Priya',21,'BBA',78),
(103,'Amit',22,'BCA',90),
(104,'Neha',20,'BCom',72),
(105,'Arjun',21,'BCA',88);
```

Step 3: Students in BCA Course

```
SELECT * FROM Students
WHERE Course = 'BCA';
```

Step 4: Students Scoring More Than 80

```
SELECT * FROM Students
WHERE Marks > 80;
```

Step 5: Students Age 21

```
SELECT * FROM Students
WHERE Age = 21;
```

Result

- The WHERE clause is used to filter records.
- It allows retrieval of specific data based on conditions.
- Different comparison operators such as =, >, <, >=, <= can be used with the WHERE clause.

Conclusion

The WHERE clause was successfully implemented to retrieve specific records based on different conditions.

practice question:-002

Aim

To study and implement the **ORDER BY clause** in SQL.

Problem Statement

Create a table named **Employees** to store employee details and use the ORDER BY clause to sort the records in ascending and descending order.

Table: Employees

EmpID	EmpName	Department	Salary	Age
201	Rahul	HR	30000	25
202	Priya	IT	50000	28
203	Amit	Finance	45000	30
204	Neha	IT	55000	27
205	Arjun	HR	32000	26

Tasks to Perform

(a) Create the Employees table.
(b) Insert the given records.
(c) Display all employees sorted by Salary in ascending order.
(d) Display all employees sorted by Salary in descending order.
(e) Display employees sorted by Department (A–Z).
(f) Display employees sorted by Age in descending order.

Solution

Step 1: Create Table

```
CREATE TABLE Employees (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    Department VARCHAR(30),
    Salary INT,
    Age INT
);
```

Step 2: Insert Records

```
INSERT INTO Employees VALUES
(201,'Rahul','HR',30000,25),
(202,'Priya','IT',50000,28),
(203,'Amit','Finance',45000,30),
(204,'Neha','IT',55000,27),
(205,'Arjun','HR',32000,26);
```

Step 3: Sort by Salary (Ascending)

```
SELECT * FROM Employees
ORDER BY Salary ASC;
```

Step 4: Sort by Salary (Descending)

```
SELECT * FROM Employees
ORDER BY Salary DESC;
```

Step 5: Sort by Department (A–Z)

```
SELECT * FROM Employees
ORDER BY Department ASC;
```

Step 6: Sort by Age (Descending)

```
SELECT * FROM Employees
ORDER BY Age DESC;
```

Result

- The ORDER BY clause is used to sort records.
- **ASC** sorts in ascending order (default).
- **DESC** sorts in descending order.
- Multiple columns can also be sorted using ORDER BY.

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Conclusion

The ORDER BY clause was successfully implemented to sort employee records in different orders based on various columns.

practice question:-003

Aim

To study and implement **AND, OR, NOT** operators in SQL using the WHERE clause.

Problem Statement

Create a table named **Employees** and use AND, OR, NOT operators to filter records based on multiple conditions.

Table: Employees

EmpID	EmpName	Department	Salary	Age
201	Rahul	HR	30000	25
202	Priya	IT	50000	28
203	Amit	Finance	45000	30
204	Neha	IT	55000	27
205	Arjun	HR	32000	26

Tasks to Perform

(a) Create the Employees table.
(b) Insert the given records.
(c) Display employees who belong to IT department AND salary is greater than 50000.
(d) Display employees who belong to HR department OR salary is greater than 50000.
(e) Display employees who are NOT from IT department.
(f) Display employees whose salary is greater than 30000 AND age is less than 30.

Solution

Step 1: Create Table

```
CREATE TABLE Employees (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    Department VARCHAR(30),
    Salary INT,
    Age INT
);
```

Step 2: Insert Records

```
INSERT INTO Employees VALUES
(201,'Rahul','HR',30000,25),
(202,'Priya','IT',50000,28),
(203,'Amit','Finance',45000,30),
(204,'Neha','IT',55000,27),
(205,'Arjun','HR',32000,26);
```

Step 3: AND Operator

```
SELECT * FROM Employees
WHERE Department = 'IT' AND Salary > 50000;
```

Step 4: OR Operator

```
SELECT * FROM Employees
WHERE Department = 'HR' OR Salary > 50000;
```

Step 5: NOT Operator

```
SELECT * FROM Employees
WHERE NOT Department = 'IT';
```

Step 6: Multiple Conditions (AND)

```
SELECT * FROM Employees
WHERE Salary > 30000 AND Age < 30;
```

Result

- AND returns records when all conditions are true.
- OR returns records when at least one condition is true.
- NOT excludes records that meet the given condition.
- These operators help apply multiple filtering conditions in SQL.

Conclusion

The AND, OR, and NOT operators were successfully implemented using the WHERE clause to filter employee records based on multiple conditions.

practice question:-003

Aim

To study and implement the **UPDATE** and **DELETE** statements in SQL.

Problem Statement

Create a table named **Employees** and perform UPDATE and DELETE operations on the records.

Table: Employees

EmpID	EmpName	Department	Salary	Age
201	Rahul	HR	30000	25
202	Priya	IT	50000	28
203	Amit	Finance	45000	30
204	Neha	IT	55000	27
205	Arjun	HR	32000	26

Tasks to Perform

(a) Create the Employees table.
(b) Insert the given records.
(c) Update the salary of employee Rahul to 35000.
(d) Update the department of employee with EmpID = 203 to "HR".
(e) Delete the employee whose EmpID = 205.
(f) Delete employees whose salary is less than 35000.
(g) Display the final table.

Solution

Step 1: Create Table

```
CREATE TABLE Employees (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    Department VARCHAR(30),
    Salary INT,
    Age INT
);
```

Step 2: Insert Records

```
INSERT INTO Employees VALUES
(201,'Rahul','HR',30000,25),
(202,'Priya','IT',50000,28),
(203,'Amit','Finance',45000,30),
(204,'Neha','IT',55000,27),
(205,'Arjun','HR',32000,26);
```

Step 3: UPDATE Salary of Rahul

```
UPDATE Employees
SET Salary = 35000
WHERE EmpName = 'Rahul';
```

Step 4: UPDATE Department of EmpID = 203

```
UPDATE Employees
SET Department = 'HR'
WHERE EmpID = 203;
```

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Step 5: DELETE Employee with EmpID = 205

```
DELETE FROM Employees
WHERE EmpID = 205;
```

Step 6: DELETE Employees with Salary < 35000

```
DELETE FROM Employees
WHERE Salary < 35000;
```

Step 7: Display Final Table

```
SELECT * FROM Employees;
```

Result

- UPDATE is used to modify existing records.
- DELETE is used to remove records from a table.
- WHERE clause is important to avoid updating or deleting all records accidentally.

Conclusion

The UPDATE and DELETE statements were successfully implemented to modify and remove records in the Employees table.

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